

Despite the promise, many companies still fall short in embracing data-driven decision making

Throughout my 15+ year of experience in corporate environments, I've observed consistent patterns,

... roadblocks that repeatedly limit progress, stifle insights and reduce the impact of analytics,

... no matter the size of the company or the enthusiasm of its people.

5+1 BARRIERS TO EFFECTIVE DATA-DRIVEN DECISION MAKING



Lack of Focus and Actionability

An inability to clearly identify key drivers, understand their root causes, or determine the most effective actions, making it challenging to address relevant issues, maximize profits, and minimize risks.



Inadequate Information Management

Have you ever witnessed people in an executive meeting presenting different numbers for the same metric? Lack of clear definitions, or inconsistent and inaccurate data, undermine trust and hinder visibility and optimal decision-making.



Fragmented Data Architecture

Having the right data in a seamless way to create reports and perform analytics productively, requires consolidated datasets that are refreshed automatically, and platforms that enable teams to collaborate.



Underutilization of Modern Tools

Adoption and effective use of AI & Analytical tools remain limited, while they are necessary to enable our people increase their productivity and unify data inputs to obtain insights.

Organizational & Cultural Barriers

...arise from centralized decision-making, resistance to change, and a lack of leadership commitment to a data-driven

Decision Execution Gap

occurs when organizations struggle to act on available insights due to misalignment with business priorities and slow decision cycles, preventing timely and effective



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Empowering Businesses with Smarter Insights.

Matt has worked with global corporations, leading consultancies, and major universities, in various analytical roles, including BI Manager, Data Analyst, Data Scientist & Data Engineer, for over 20 years in total. He's been providing value to clients by developing automated reports, building data warehousing infrastructure, and empowering their analytical transformation; helping them foster a data-driven culture and leverage AI & advanced analytical tools.

20 +

Years of Data and Analytics experience

BI (automated) Reports
Data Lakehouses
Advanced Analytics & AI

30 +

Companies worked with in various sectors

Banking
Supply Chain
Pharma
Energy
Chemicals
FMCG
...

450 +

Power BI reports & Analytics deliverables

2000 +

Project Days in BI Development and Data Engineering

20 +

Professional Certifications



... and more

Some of the organizations Matt has worked with:



Lack of Focus and Actionability

Sales Report

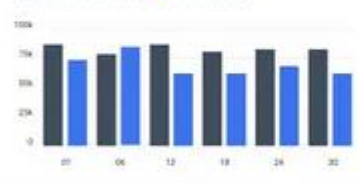
Cost
US\$ 32,900
+1.39%

Goal completions
56,963
+3.49%

Cost per click
US\$ 2,50
+0.13%

Campaign conversions

● Transaction revenue ● Transactions



Breakdown by Country

Country	Spent	Reach	Clicks
Germany	\$20,064	9,942	8,932
Denmark	\$18,052	7,753	6,945
Austria	\$17,032	6,934	5,578

Reach over time

○ Reach



Lack of focus and actionability refers to the tendency of business reports to present data that is broad, generic, or disconnected from the decisions that leaders need to make.

Instead of highlighting the specific drivers of performance or providing clear, actionable insights, these reports often focus on surface-level metrics—like sales or costs—without deeper context or analysis.

For example, in the report above, the areas that need improvement are not evident, the reasons why are missing, and there is not a clear path to action.



Decision-makers are overwhelmed with data but lack clarity on what truly matters



Key performance drivers are either unknown or poorly understood, making it hard to link outcomes (e.g., sales growth or cost increases) to specific internal or external factors



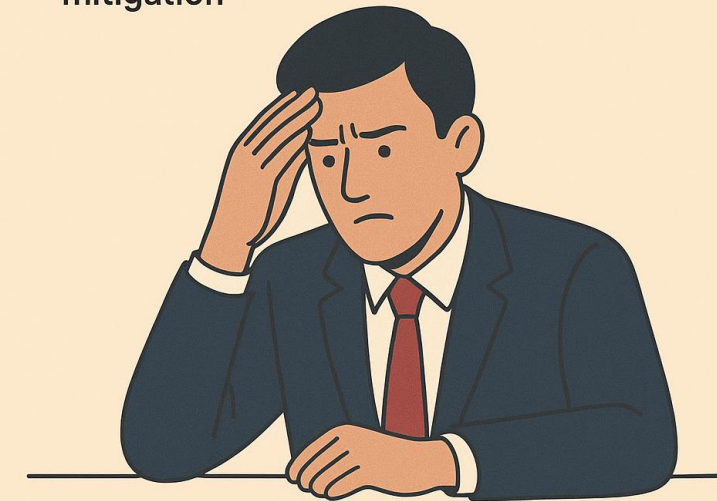
Reports highlight symptoms, not causes, limiting the ability to identify the root of problems or opportunities



There's little guidance on what actions to take, which leads to indecision, inefficiencies

As a result, leaders struggle to:

- Prioritize effectively,
- Align teams around the most impactful levers of performance
- React swiftly to changing conditions, and
- Drive sustainable growth and risk mitigation



Ultimately, a lack of focus and actionability undermines the very purpose of reporting: to enable better, faster, and more informed decision-making.

...can be resolved with properly designed KPI's & Analytics

Knowing the volume of your sales, or the numbers in the 'bottom line' is important but cannot help you decide what to do to increase them.

You need to focus on what drives sales, profits and value...



KPIs with clear targets, and a clear view on when targets are not met.

Then leverage analytics to understand why,



define proactive action plans to maximize performance,

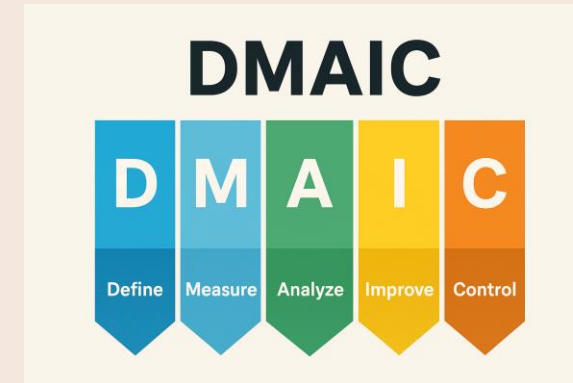
and put in place controls to trigger them.



Large companies have both business experts within the COO, CFO & CFO mandates, and access to management consulting firms, to determine KPI's and root causes for them.

SMEs on the other hand can either:

- ① **Try to adapt what works within their business models**
- ② **Use a framework to help them identify what to measure and analyze information**



When I was leading the BI & Big Data Team at PeopleCert, we worked together with a management consultant to model the entire set of business processes, and create a set of interconnected Power BI reports, for the senior stakeholders to review and decide how to move the business forward. We even made a summary video of the project:

[!\[\]\(19d44b37fb4fa155bf9d60c77a3d3cb2_img.jpg\) Management Dashboard with System Dynamics](#)

- ③ **Use a structured problem-solving approach to identify & validate potential drivers.**

For example, MECE logical trees, popularized by an ex-McKinsey employee, can be used to break complex problems into clear, non-overlapping parts that cover all possibilities, helping structure analysis, avoid gaps, and initiate analytical investigations, to support data-driven decisions.

Data quality and governance gaps are undermining organizational decision-making.

Large enterprises struggle with entrenched data silos, while smaller businesses rely on ad-hoc, uncontrolled processes.

Inconsistent or inaccurate data erodes trust and limits actionable insights, while technical challenges in integrating multiple systems create fragmented processes and poor visibility.

Without robust data governance, organizations risk inefficiency, missed opportunities, and strategic missteps, turning data from a valuable asset into a critical liability.

Organizations today face critical challenges in data quality and governance that directly affect their strategic decision-making.

In large enterprises, entrenched data silos prevent a cohesive view of operations, while smaller businesses often rely on ad-hoc, uncontrolled processes that leave data vulnerable to errors. Across the board, inconsistent, incomplete, or inaccurate data erodes trust and limits the reliability of analysis, undermining confidence in key business insights.

The consequence is a lack of operational transparency and limited visibility into core business performance, which in turn restricts the ability to make informed, timely, and optimal decisions.

Without a robust approach to data governance and quality management, organizations risk strategic missteps, inefficiencies, and missed opportunities, making data not an asset but a liability in driving growth and competitive advantage.

The 7 Must-Have Foundations for Modern Data and Analytics Governance



① Data Governance Framework

A robust data governance framework is essential for consistent, reliable, and trustworthy data.

This involves clearly defining data ownership, responsibilities, and policies across the organization.

By setting standards for data creation, storage, and usage, companies can reduce inconsistencies and prevent data silos.

A governance council or steering committee ensures compliance, monitors performance, and addresses emerging data issues proactively.

Formal governance processes also make it easier to scale initiatives and maintain data quality as the organization grows.

② Standardize Processes and Documentation

Ad-hoc processes introduce variability, errors, and inconsistencies.

Standardizing workflows, documenting data definitions, and formalizing business rules create clarity and consistency.

Clear documentation ensures employees know how to handle data correctly and maintain best practices.

Training teams on these standards fosters accountability and reduces reliance on informal knowledge, leading to repeatable and reliable outcomes.

③ Formalize Data Quality Management Practices

High-quality data requires proactive monitoring, validation, and cleansing.

Regularly profiling data to detect errors, missing values, or inconsistencies prevents unreliable insights and supports confident decision-making.

Automation can flag anomalies in real time, while KPIs track improvements over time.

Embedding these practices into daily workflows through automatic reports and AI tools, reduces costly mistakes and builds trust in analytical outputs.

④ Leverage Technology and Integration Tools

Technical solutions are critical for harmonizing data across multiple systems.

ETL/ELT pipelines, Master Data Management (MDM), and APIs help ensure data is accurate, consistent, and integrated.

These tools reduce manual effort, streamline data flows, and create a single source of truth, enabling faster, more reliable decision-making.

Choosing scalable, flexible technologies also future-proofs the organization as data sources and systems evolve.

⑤ Break Down Data Silos

Data silos prevent organizations from gaining a holistic view of operations, leading to inefficiencies and inconsistent decision-making.

Breaking down these silos requires cross-department collaboration and shared ownership of data.

Implementing centralized data platforms, such as data warehouses or lakes, and standardizing data definitions ensures that all teams operate using the same "data language."

This alignment reduces redundancy, improves transparency, and enables better analytical insights.

⑥ Implement Automated Data Validation and Monitoring

Technical controls such as automated validation rules, real-time monitoring, and anomaly detection are essential to maintain data integrity, accuracy, and timeliness.

Systems can flag missing, inconsistent, or duplicate data before it propagates through analytics or operational processes.

Advanced monitoring dashboards provide visibility into data quality metrics across all sources, enabling proactive issue resolution.

Integrating these tools with data pipelines ensures continuous oversight and reduces the risk of errors impacting decision-making.

A fragmented Data Architecture

Fragmented data architectures prevent organizations from accessing accurate, consistent, and timely information.

Disconnected systems and manual data consolidation lead to reporting delays, inconsistent metrics, and unreliable analytics.

This undermines confidence in decision-making, increases operational inefficiency, and limits visibility across the business.

Without a unified, well-governed data foundation, leaders face greater risk of misinformed strategies and missed opportunities.

A fragmented data architecture is one of the most pervasive barriers to achieving reliable, data-driven decision-making in modern organizations.

When data resides in disconnected systems, formats, or platforms, it becomes difficult to access, reconcile, and trust. Teams spend significant time manually consolidating datasets, often leading to inconsistencies, duplicated effort, and delays in reporting.

The lack of integration prevents the creation of a single, accurate version of the truth—causing discrepancies in metrics, dashboards, and analytics outputs.

As a result, business leaders make decisions based on incomplete or outdated insights, undermining confidence in data and eroding organizational trust.

Beyond analytics, fragmented architectures also increase operational costs, hinder collaboration, and make it challenging to implement governance and compliance controls.

Ultimately, without a cohesive and unified data foundation, the business struggles to achieve agility, efficiency, and strategic clarity in a data-driven world.



Requires an effective set of technologies working together,

① Adopt a Unified Data Platform

Fragmented systems can be consolidated by implementing a unified data platform, such as a modern data warehouse, data lakehouse, or cloud-native analytics environment (e.g., Databricks, MS Fabric).

These platforms centralize data storage, simplify access, and enable teams to work from a single, consistent source of truth.

A unified platform also supports scalability and performance, allowing organizations to integrate structured, semi-structured, and unstructured data efficiently.

This eliminates duplication and streamlines analytical workloads across departments.



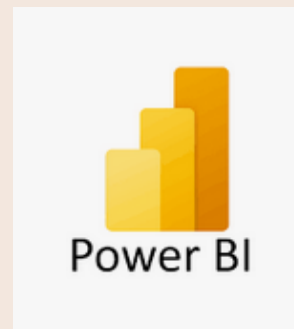
② Implement Automated Data Integration Pipelines

Manual data aggregation is prone to delays and inconsistencies.

By deploying ELT (Extract – Load – Transform) pipelines, organizations can automate the ingestion of data from multiple systems.

Automation ensures that datasets are refreshed regularly and consistently, reducing latency and the risk of stale information.

Well-designed integration pipelines enable near real-time reporting and analytics, empowering teams with up-to-date insights.



③ Standardize Data Models and Definitions

Inconsistent data structures and definitions are a common cause of fragmentation.

Establishing a standardized data model—with clearly defined dimensions, metrics, and naming conventions—ensures uniformity across systems and reports.

This alignment allows teams to interpret data consistently, reducing confusion and improving collaboration between business units, analysts, and data engineers.



④ Leverage APIs and Data Virtualization

Instead of relying solely on physical consolidation, organizations can use APIs and data virtualization to provide seamless access to distributed data.

This approach allows teams to query and combine data across different systems in real time without complex migrations.

Data virtualization creates a logical data layer that provides unified access and visibility, improving flexibility while maintaining governance and security controls.

⑤ Enhance Collaboration Through Shared Data Workspaces

Effective analytics requires teams to work together using common tools and interfaces.

Implementing collaborative data environments—such as shared BI dashboards, data catalogs, or cloud-based notebooks—fosters transparency and shared understanding.

These platforms also allow business users and data professionals to co-develop reports, annotate findings, and manage data assets collaboratively, reducing duplication of effort and ensuring consistent use of approved data sources.

⑥ Follow solid Data Architecture Management Practices

According to DAMA DMBOK, Data Architecture Management provides the structural foundation for aligning data strategy with business goals, ensuring that data assets are organized, standardized, and integrated to support enterprise information needs.

Applying these principles means designing an architecture that is metadata-driven, business-aligned, and governed through clear standards and stewardship.

It emphasizes the importance of defining data domains, relationships, and lineage—ensuring that the architecture supports not just analytics, but also compliance, interoperability, and long-term scalability.

By grounding data architecture in best practices, organizations can move from fragmented systems toward a cohesive, well-governed, and future-ready data ecosystem.

Many organizations underutilize modern data, analytics, and AI tools, limiting their ability to make fast, informed decisions.

Despite the availability of advanced technologies, adoption remains low due to limited data literacy and reliance on technical specialists.

This dependence slows decision-making and reduces agility, while business users often lack the training to apply analytics effectively.

As a result, companies fail to unlock the full value of their data investments and struggle to build a truly data-driven culture that supports smarter decision-making.

Many organizations today struggle with the underutilization of modern tools in data, analytics, and reporting, despite the widespread availability of advanced technologies such as AI, machine learning, and automation platforms.

These tools have the potential to dramatically improve decision-making speed, accuracy, and strategic foresight.

However, adoption often remains limited due to a shortage of employees with strong data literacy and the technical skills needed to use these tools effectively.

As a result, businesses become overly dependent on data specialists, creating bottlenecks that delay insights and slow decision-making. Without sufficient training and empowerment, business users are unable to translate analytical capabilities into practical, day-to-day value.

This gap between technological potential and actual utilization prevents organizations from realizing the full return on their data investments and from developing a truly data-driven culture.

① Enhance Data Literacy Across the Organization

Develop structured training programs to improve employees' understanding of data concepts, analytical thinking, and visualization tools.

Building data literacy at all level - especially among business users empowers teams to interpret insights independently and make informed decisions without relying solely on specialists.

② Simplify Access to Analytics Platforms & Tools

Adopt user-friendly, self-service analytics platforms that allow non-technical users to explore data, generate reports, and build dashboards easily.

Reducing technical complexity increases engagement and ensures that insights can be produced quickly and collaboratively across departments.

③ Integrate Automation and AI into Business Workflows

Embed automation and AI capabilities directly into business processes to enhance productivity and decision-making.

Automating data preparation, reporting, and forecasting tasks reduces manual effort, allowing teams to focus on higher-value analysis and strategic actions.

④ Provide Continuous Training and Upskilling

Offer ongoing professional development programs tailored to different user groups—from data novices to power users. Encourage certifications, workshops, and mentorship initiatives to keep employees aligned with evolving technologies and best practices in analytics and AI.

⑤ Foster Collaboration Between IT and Business Teams

Bridge the gap between technical experts and business users through cross-functional collaboration. Establish “data champions” or “analytics ambassadors” within departments to facilitate communication, promote tool adoption, and ensure analytics align with real business needs.

⑥ Establish a Data Enablement Strategy

Create a dedicated data enablement function that focuses on maximizing the use and value of modern analytics tools.

This includes defining tool usage standards, ensuring integration with existing systems, and promoting a culture of experimentation and data-driven innovation.

A data-driven culture has tremendous value on itself

The social interactions and norms within an organization, and in particular how decisions are made, are the main aspects of a data-driven culture - or the lack thereof.

Gut feeling is one way to make decisions, but is not very effective, and neither is scalable.

Culture is not about what we think we do; it is about what we actually do. The type of behaviors that are expected, rewarded and reinforced in a group is what establishes a culture.

Hence the saying, 'culture eats strategy for breakfast'.

Even if we have the best of intentions, and have thought deeply about how to achieve the next steps...

... norms do not change overnight.

The value of the data depends solely on whether they are used in a way that creates value.

And that's all about whether the decisions made are realistic and justified or are just about nodding in agreement to the predispositions of authority.

Smaller businesses rely heavily on a few people.

Decisions remain centralized, limiting growth.

In larger companies, resistance to change is common, while overreliance on intuition over data is not uncommon either.

Lack of leadership commitment to foster a data-driven culture, and misaligned incentives between the various teams within the organization, lead to suboptimal and uncoordinated actions.

Without a data-driven culture, proactivity and scalability suffer.

① Executive Sponsorship

A data driven culture needs have 'executive buy-in'. If leadership is not aware of the value gains and the problems that need to be solved, then even the most persistent efforts to build an effective data-driven culture will fail.

The executive sponsor, is a leader that can align (and rally) the people towards a common goal. The benefits that employees gain when the adopt one set of practices instead of another, are defined by leadership. This plays a pivotal role on culture evolution, and the leader must demonstrate that data-driven decision is placed high in the set of company values.

The leader can also secure the necessary resources, such as the development of data & analytical assets necessary for generating value from data. They can provide funding and time for training and foster a 'safe space' for experimentation.

All these aspects are critical for a data-driven culture to thrive.

② Pluralism: various roles and tools

An organization can have different modes of developing analytical value from data.

Corporate BI has been a centralized, IT driven way of creating reports and analytical data assets for decades, but the requirements for agility and speed has led to the adoption of flexible **business-led, self-service analytics** alongside the expensive data ware(-lake)houses and automatic reports displaying metrics in a consistent way, as per business definitions.

The agility and emergence of novel insights of self-service analytics, are now elevated even more with modern **AI tools** that can help analysts reach value even faster.

So, they are two forces at play: creativity and the need for consistency that must interwind together for value to be maximized: **'discipline at the core & flexibility at the edge'**, as per a famous quote used from many Microsoft employees.

③ Community of practice

An informal –and extensive- team of professionals within the organization that communicate and work together to develop assets, ways of working and value for the common good of the company, is a community of practice.

The community plays a pivotal role on the development of culture. If no such community exist, then the culture cannot be fostered in an effective way.

And while the technical tools that facilitate communication (chat, forums, e-mail lists, video channels, and a perhaps a central portal to host all those assets) are commonplace today, there is always the need for **champions** – people who promote and demonstrate the values of data-driven decisions and ways of working , and a **Center of Excellence** – a group of people who can guide the rest of the community through their expertise on either technical best practices on managing and analyzing data or making effective data-driven decisions.

4 Motivation

Motivation and encouragement to use data-driven practices is also necessary. Intrinsic motivation – the joy we derive from doing the work itself in a creative way- is always superior to extrinsic motivation – bonuses, promotions and anything we are promised to gain if we perform it well.

Of course both are necessary, and it is important to link extrinsic motivators to following the values of the culture we want to build, otherwise those discrepancies can act as de-motivators. However, the creativity, sense of belonging to the community and mental health benefits that derive intrinsically can play the most drastic role in happiness and productivity.

Judging from myself, in various phases in my career, I gained more money in a project... not always it did translate on me being happier or feeling 'unstoppable' in regard to my creativity and productivity.

5 Training

The value of the availability of the necessary tools, and the necessary training to use them is apparent. Even a community of motivated individuals that receive guidance from the best experts cannot by themselves take the initiatives they need to take to structure their thinking and generate the insights they need to realize value from information.

Here the role of the executive sponsor is again evident, as (s)he should ensure all the necessary resources are in place.

6 Measuring Success

All the previous strategic and tactical steps – and the subsequent actions an organization needs to take- should always come along with the question: "how do we know it's actually working?". This is the only way we can progress towards the goal of building a data-driven culture.

A simple way would be to look at the number of people who are actively involved in ways that align well with the values of our data-driven culture. Of course that does not measure success – it measures usage. A more subtle way would be to evaluate whether the analytics that are required for people who need to make decisions to be able to make them in a timely and data informed manner, are there – or are evolving towards the desired state.

Finally, the most important measure – and perhaps the most difficult to attain- is value. The end value of the decisions made, and how they are better when they are based on data and information, instead of intuition, would be the measure of success with the most importance. It is of course difficult to attribute an increase in profits to specific analytical initiatives.

However, the importance of culture is that encompasses the entire organization. And no one can deny that culture, not strategy or indeed tactics, which are eaten for breakfast ☺ , provide the most impact on achieving every one of these goals.

A decision-execution gap occurs when organizations fail to act quickly on data-driven insights due to misalignment with business priorities and slow decision processes.

Valuable insights often remain unused, leading to missed opportunities and reduced competitiveness. Inefficient communication, unclear ownership, and fragmented workflows further delay execution.

Closing this gap requires aligning strategy with analytics, streamlining decision-making, and empowering teams to act on insights rapidly to drive timely, effective business outcomes.



① Align Insights with Business Priorities

Ensure that analytics and reporting focus on the most critical business objectives.

This aligns well with the principle of a focused approach to what matters most.

Prioritize insights that directly impact strategic goals and operational performance so that decision-makers can act quickly on what matters most.

② Streamline Decision-Making Processes

Reduce bureaucratic bottlenecks by clarifying roles, responsibilities, and approval workflows.

Empower teams with clear authority to act on insights without unnecessary delays, enabling faster, more agile responses.

Embed data and analytics into day-to-day processes through dashboards, alerts, and automated recommendations.

This ensures that insights are visible and actionable in the context where decisions are made, bridging the gap between analysis and execution.

③ Integrate AI into Operational Workflows

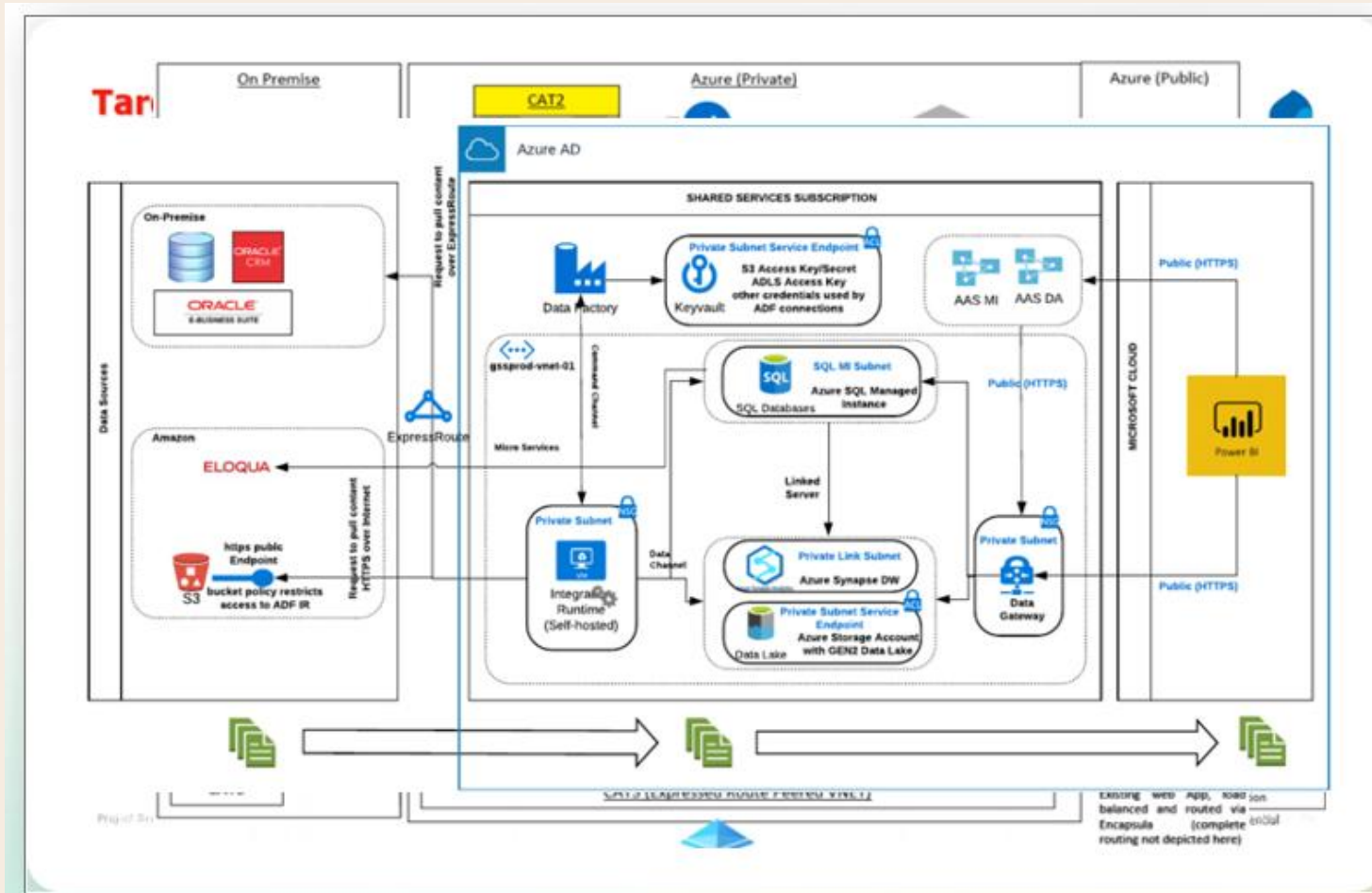
Modern AI agents and analytics platforms can embed actionable insights directly into operational workflows, making it easier for teams to act in real time.

For example, AI-driven dashboards can prioritize alerts based on business impact, automatically generate recommended actions, or even trigger workflow tasks without manual intervention.

Intelligent agents can summarize large volumes of data, surface key trends, and provide contextual guidance tailored to the user's role, ensuring decisions are informed and timely.

By integrating AI into routine processes—whether in sales, supply chain, marketing, or customer service—organizations reduce reliance on manual interpretation, accelerate execution, and close the gap between insight generation and decision-making.

To ensure you can generate value from all the modern technological capabilities, and go from low analytics engagement, with fragmented, 'ad-hoc' reporting based on low quality data, to high-value decisions powered by actionable insights and AI-powered solutions that increase productivity and profits



You need to follow the right path...

Let's discuss how Data & Analytics can help your business thrive

KEDROS ANALYTICS
Empowering Businesses with Smarter Insights.

Schedule a Free 30'
Discovery Call



...or drop me a mail at free.audit@kedrosanalytics.com